

Features

- Linear current limitation
- Thermal shutdown
- Short circuit protection
- Integrated clamp
- Low current drawn from input pin
- Diagnostic feedback through input pin
- ESD protection
- Direct access to the gate of the Power MOSFET (analog driving)
- Compatible with standard Power MOSFET
- Automotive qualified (AEC-Q100 Grade 1)



Description

The HELS121 is intended for replacement of standard Power MOSFET.

Built-in thermal shutdown, linear current limitation and over voltage clamp protect the chip in harsh environments.

Fault feedback can be detected by monitoring voltage at the input pin.

Type	R _{DS(on)}	I _{lim}	V _{clamp}
HELS121-SOT223	120mΩ	3.5A	40V

Part number	Package	Marking	Packing
HELS121RCAG	SOT223	HELS121	2500 / Tape & Reel

**Not sure if you got the latest data sheet or not. Please kindly contact the sales of KKChips to confirm. Thanks.*

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1 Block diagram and pin description

1.1 Block diagram

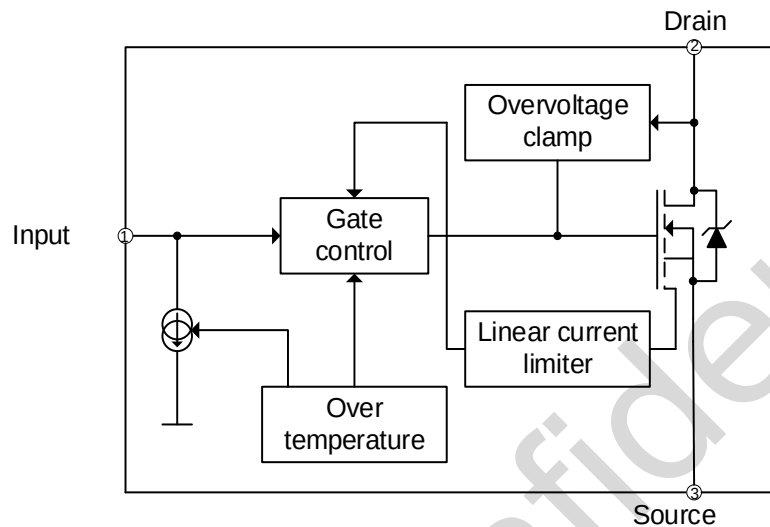


Figure 1-1 Block diagram

SOT223 Package

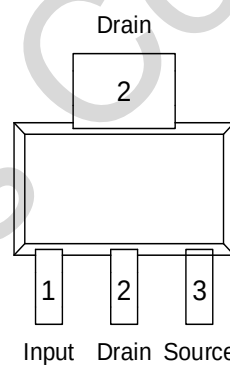


Figure 1-2 Configuration diagram (top view)

2 Electrical specifications

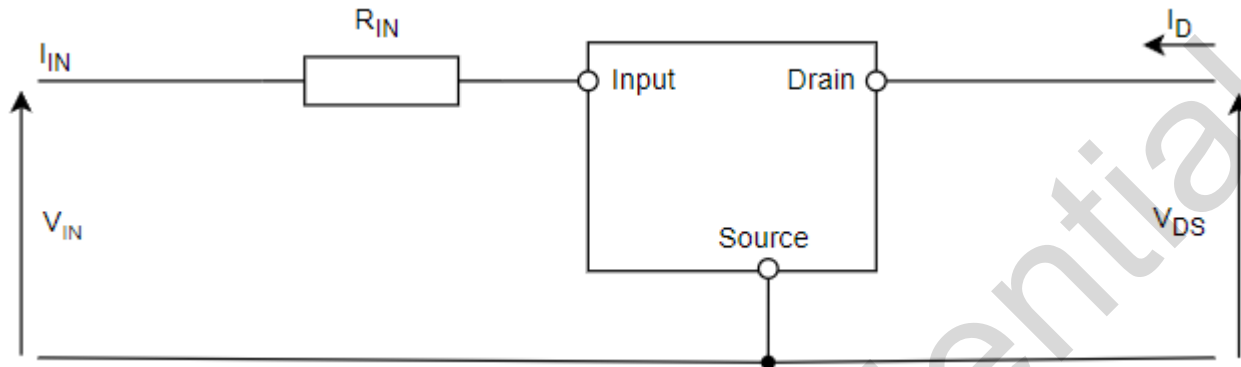


Figure 2-1 Current and voltage conventions

2.1 Absolute maximum ratings

Table 2-1 Absolute maximum ratings

Symbol	Parameter	Value	Unit
V_{DS}	Drain-Source Voltage ($V_{IN} = 0\text{ V}$)	Internally clamped	V
V_{IN}	Input voltage	Internally clamped	V
I_{IN}	Input current	+/- 20	mA
$R_{IN\text{ MIN}}$	Minimum input series impedance	220	Ω
I_D	Drain current	Internally limited	A
I_R	Reverse DC output current	-5.5	A
V_{ESD1}	Electrostatic discharge ($R = 1.5\text{ K}\Omega$, $C = 100\text{ pF}$)	4000	V
P_{tot}	Total dissipation at $T_C = 25^\circ\text{C}$	7	W
T_J	Operating junction temperature	Internally limited	$^\circ\text{C}$
T_C	Case operating temperature	Internally limited	$^\circ\text{C}$
T_{stg}	Storage temperature	-55 to 150	$^\circ\text{C}$

2.2 Thermal data

Table 2-2 Thermal data

Symbol	Parameter	Value	Unit
$R_{thj\text{-case}}$	Thermal resistance junction-case max	18	$^\circ\text{C/W}$
$R_{thj\text{-amb}}$	Thermal resistance junction-ambient max	70 ⁽¹⁾	$^\circ\text{C/W}$

1. When mounted on a standard single-sided FR4 board with 50 mm² of Cu (at least 35 μm thick)

connected to all DRAIN pins.

2.3 Electrical characteristics

$-40^{\circ}\text{C} < T_j < 150^{\circ}\text{C}$, unless otherwise specified.

Table 2-3 Off

Symbol	Parameter	Condition	Min	Typ	Max	Unit
V_{CLAMP}	Drain-source clamp voltage	$V_{IN} = 0\text{ V}; I_D = 1.5\text{ A}$	40	-	55	V
V_{CLTH}	Drain-source clamp threshold voltage	$V_{IN} = 0\text{ V}; I_D = 2\text{ mA}$	36	-	-	V
V_{INTH}	Input threshold voltage	$V_{DS} = V_{IN}; I_D = 1\text{ mA}$	0.5	1.17	2.5	V
I_{ISS}	Supply current from input pin	$V_{DS} = 0\text{ V}; V_{IN} = 5\text{ V}$	-	65	150	μA
V_{INCL}	Input-source clamp voltage	$I_{IN} = 1\text{ mA}$	5.6	6	7	V
		$I_{IN} = -1\text{ mA}$	-1.0	-	-0.3	V
I_{DSS}	Zero input voltage drain current ($V_{IN} = 0\text{ V}$)	$V_{DS} = 13\text{ V}; V_{IN} = 0\text{ V}; T_j = 25^{\circ}\text{C}$	-	-	10	μA
		$V_{DS} = 25\text{ V}; V_{IN} = 0\text{ V}$	-	-	10	μA

Table 2-4 On

Symbol	Parameter	Condition	Min	Typ	Max	Unit
$R_{DS(on)}$	Static drain-source on resistance	$V_{IN} = 5\text{ V}; I_D = 1.5\text{ A}; T_j = 25^{\circ}\text{C}$	-	-	120	m Ω
		$V_{IN} = 5\text{ V}; I_D = 1.5\text{ A}$	-	-	240	m Ω

Table 2-5 Dynamic ($T_j = 25^{\circ}\text{C}$, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
$g_{fs}^{(1)}$	Forward trans conductance	$V_{DD} = 13\text{ V}; I_D = 1.5\text{ A}$	-	5.0	-	S
C_{OSS}	Output capacitance	$V_{DS} = 13\text{ V}; f = 1\text{ MHz}; V_{IN} = 0\text{ V}$	-	150	-	pF

Table 2-6 Switching ($T_j = 25^{\circ}\text{C}$, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
$t_{d(on)}$	Turn-on delay time	$V_{DD} = 15\text{ V}; I_D = 1.5\text{ A}; V_{gen} = 5\text{ V}; R_{gen} = R_{IN\text{ MIN}} = 220\text{ }\Omega$ (see Figure 2-2)	-	90	300	ns
t_r	Rise time		-	250	750	ns
$t_{d(off)}$	Turn-off delay time		-	450	1350	ns
t_f	Fall time		-	250	750	ns
$t_{d(on)}$	Turn-on delay time	$V_{DD} = 15\text{ V}; I_D = 1.5\text{ A}; V_{gen} = 5\text{ V}; R_{gen} = 2.2\text{ K}\Omega$	-	0.45	1.35	μs
t_r	Rise time		-	2.5	7.5	μs

Electrical specifications

td(off)	Turn-off delay time	(see Figure 2-2)	-	3.3	10	μs
tf	Fall time		-	2	6	μs
(di/dt)on	Turn-on current slope	VDD =15V; ID =1.5A; Vgen =5V; Rgen =RIN MIN = 220Ω	-	4	8	A/μs
Qi	Total input charge	VDD =12V; ID = 1.5 A; VIN =5V; Igen =2.13mA (see Figure 2-5)	-	10	-	nC

Table 2-7 Source drain diode (Tj = 25°C, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
VSD ⁽¹⁾	Forward on voltage	ISD = 1.5 A; VIN =0V	-	0.8	-	V
t _{rr}	Reverse recovery time	ISD = 1.5 A; di/dt = 12 A/μs; VDD =30V; L=200μH (see Figure 2-3)	-	107	-	ns
Q _{rr}	Reverse recovery charge		-	37	-	μC
I _{RRM}	Reverse recovery current		-	0.7	-	A

Table 2-8 Protections (-40°C < Tj < 150°C, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
Ilim	Current limitation range	VIN =5 V; VDS = 13 V;	3.5	5	7	A
tdlim	Step response current limit	VIN =5 V; VDS = 13 V	-	22	-	μs
Tjsh	Over temperature shutdown		150	175	200	°C
Tjrs	Over temperature reset		135	165	-	°C
Igf	Fault sink current	VIN =5 V; VDS = 13 V; Tj = Tjsh	10	15	20	mA
Eas	Single pulse avalanche energy	Starting Tj =25°C; VDD =24V; VIN =5V; Rgen =RIN MIN =220Ω; L=24 mH (see Figure 2-4 and Figure 2-6)	100	-	-	mJ

1. Pulsed: Pulse duration = 300 μs, duty cycle 1.5%

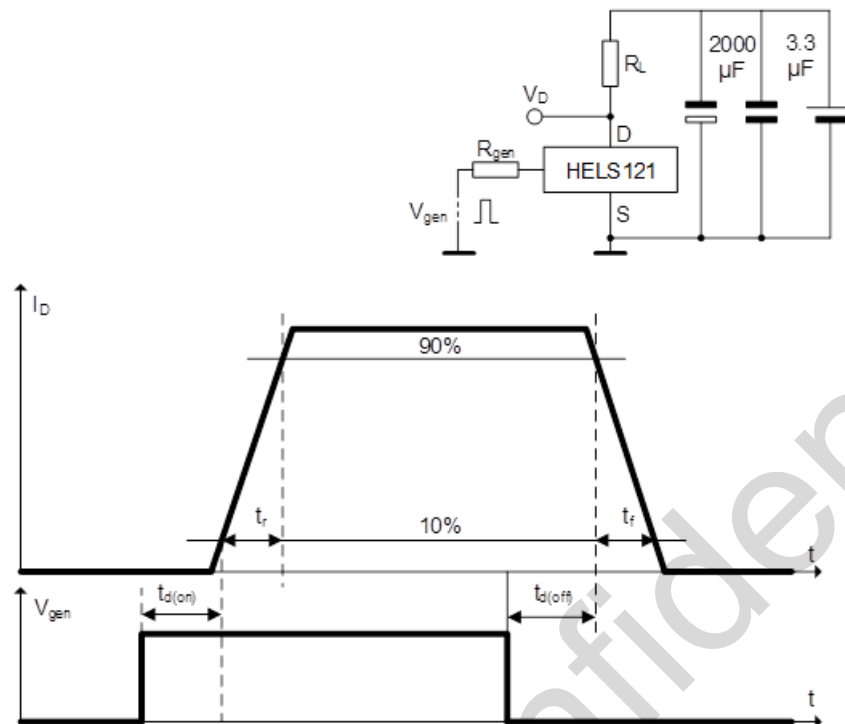


Figure 2-2 Switching time test circuit for resistive load

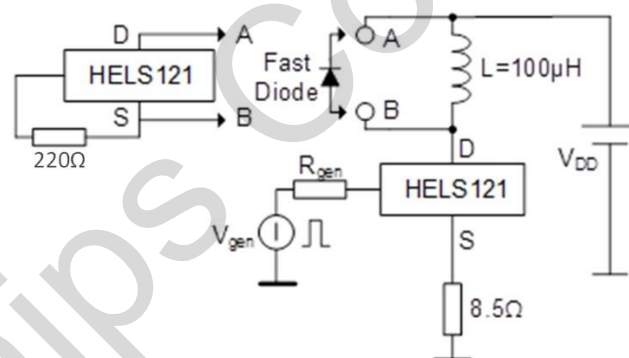


Figure 2-3 Test circuit for diode recovery times

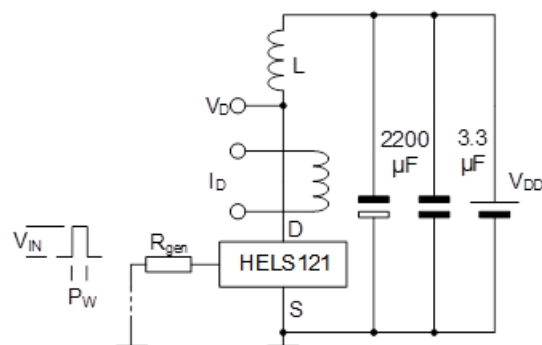


Figure 2-4 Unclamped inductive load test circuits

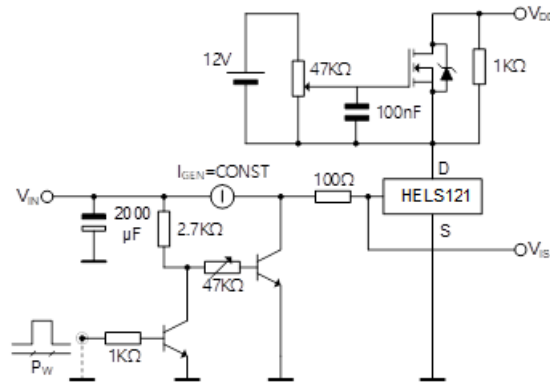


Figure 2-5 Input charge test circuit

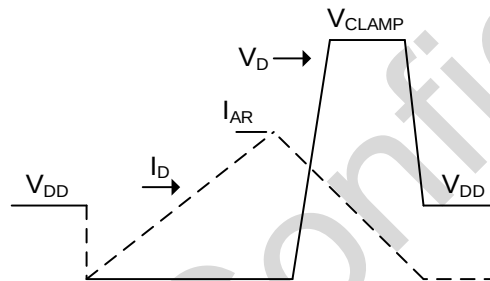


Figure 2-6 Unclamped inductive wave forms

2.4 Electrical characteristics curves

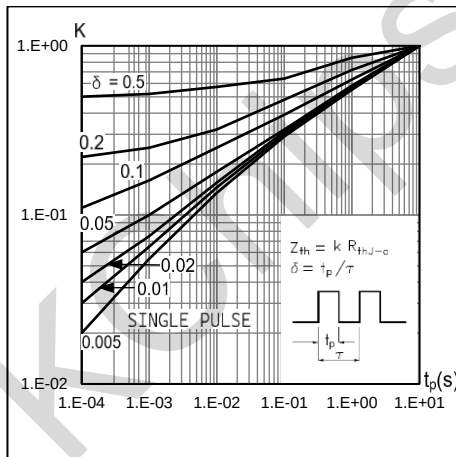


Figure 2-7 Thermal impedance for SOT223

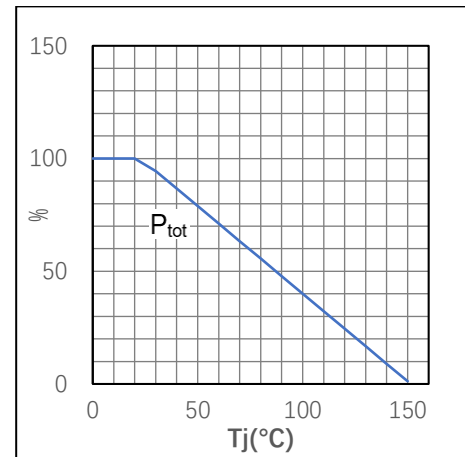


Figure 2-8 Derating curve

Electrical specifications

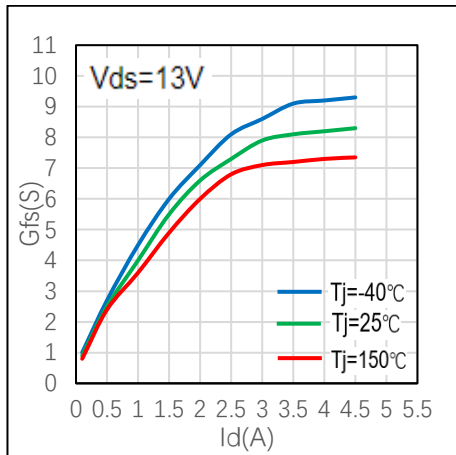


Figure 2-9 Trans conductance

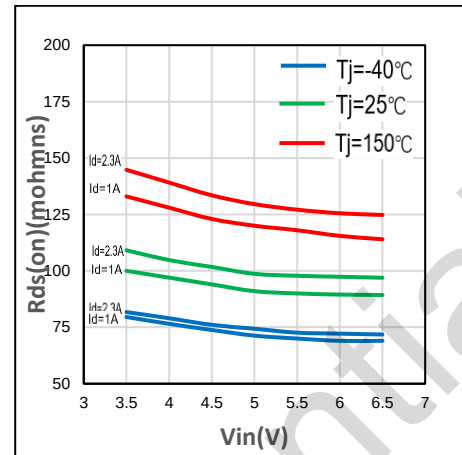


Figure 2-10 Static drain-source on resistance vs input voltage (part 1/2)

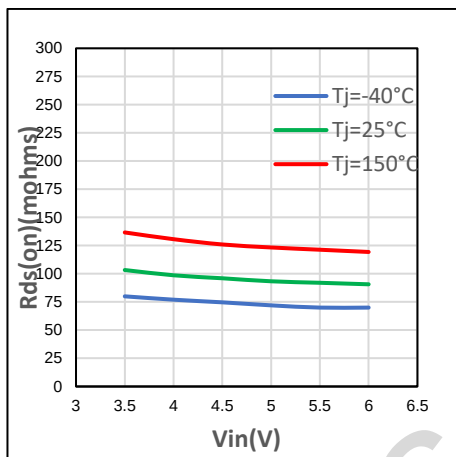


Figure 2-11 Static drain-source on resistance vs input voltage (part 2/2)

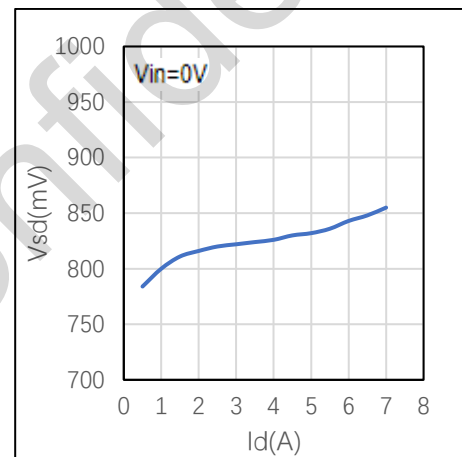


Figure 2-12 Source-drain diode forward characteristics

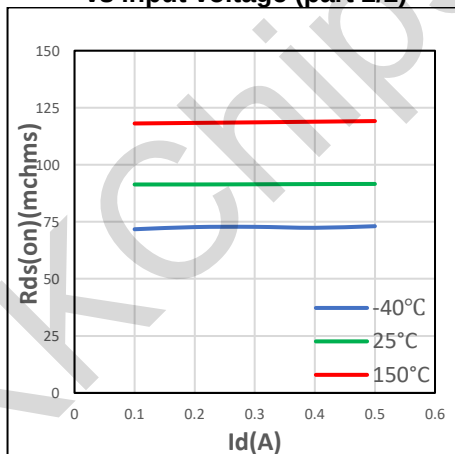


Figure 2-13 Static drain source on resistance

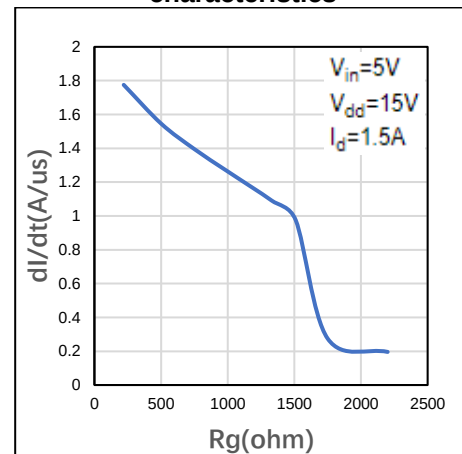


Figure 2-14 Turn-on current slope (part 1/2)

Electrical specifications

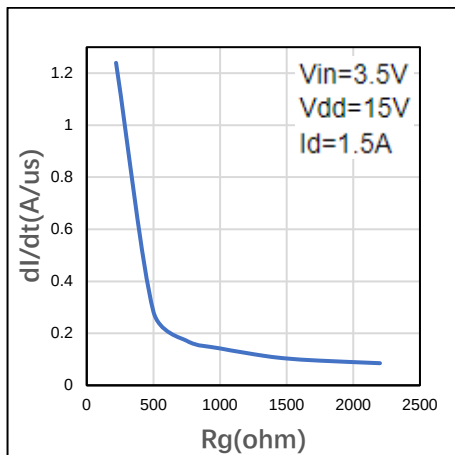


Figure 2-15 Turn-on current slope (part 2/2)

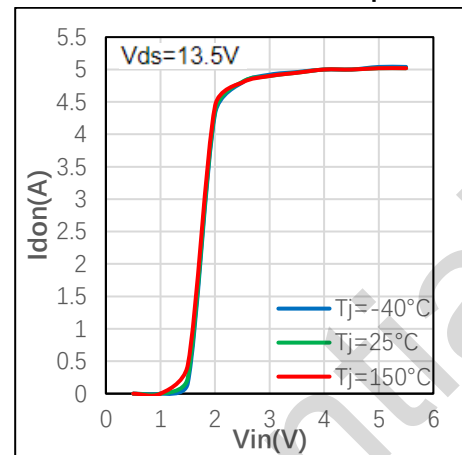


Figure 2-16 Transfer characteristics

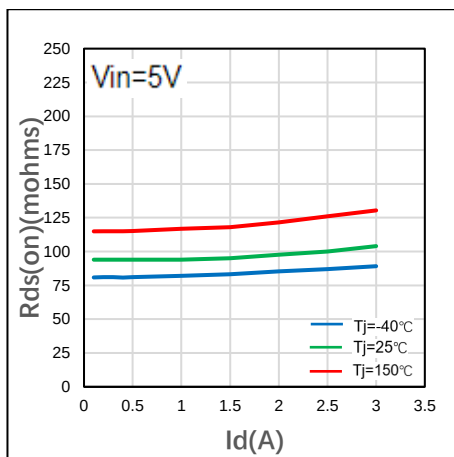


Figure 2-17 Static drain-source on resistance vs Id

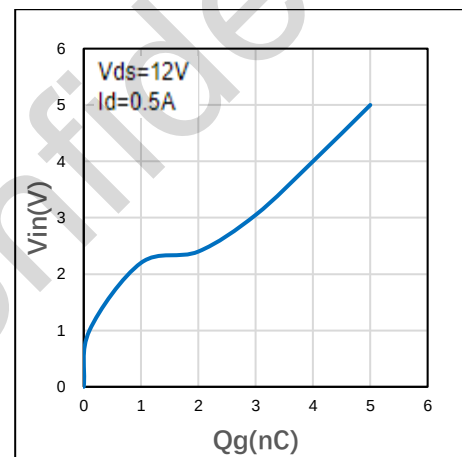


Figure 2-18 Input voltage vs input charge

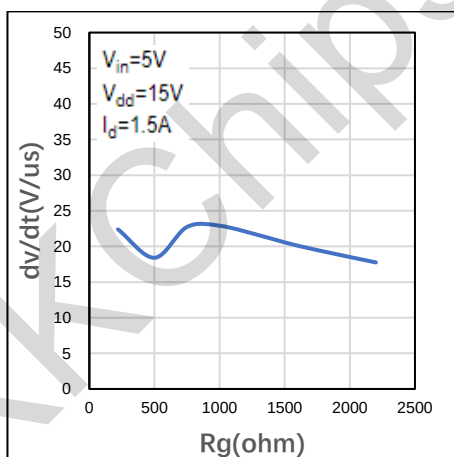


Figure 2-19 Turn-off drain source voltage slope(part1/2)

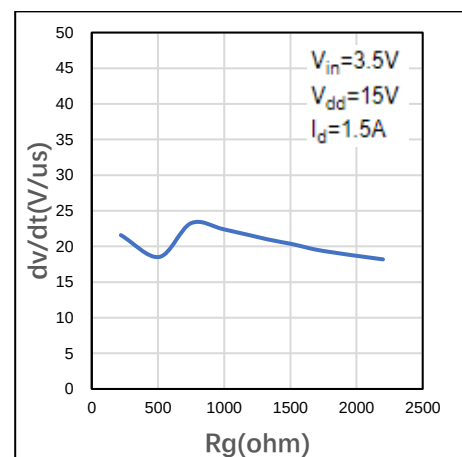


Figure 2-20 Turn-off drain source voltage slope (part 2/2)

Electrical specifications

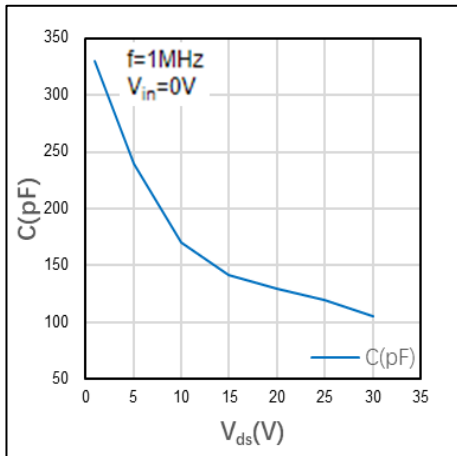


Figure 2-21 Capacitance variations

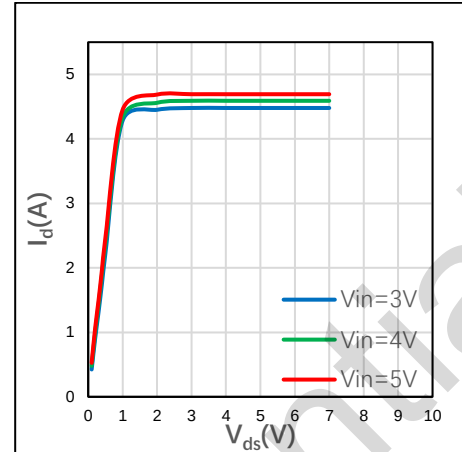


Figure 2-22 Output characteristics

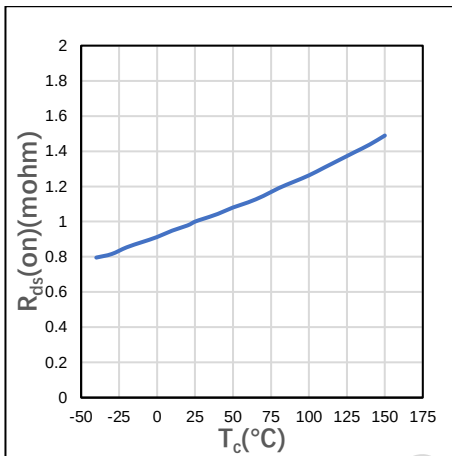


Figure 2-23 Normalized on resistance vs temperature

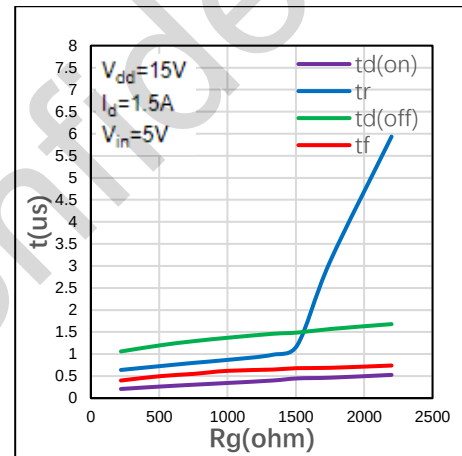


Figure 2-24 Switching time resistive load (part 1/2)

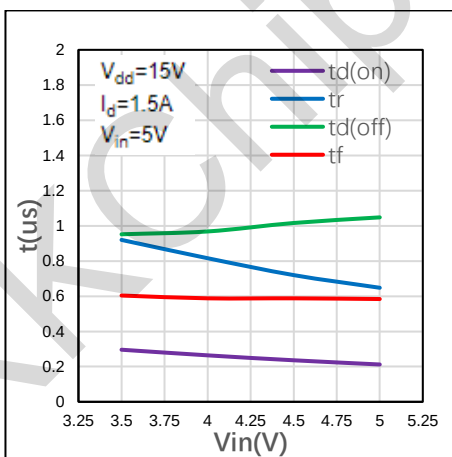


Figure 2-25 Switching time resistive load (part 2/2)

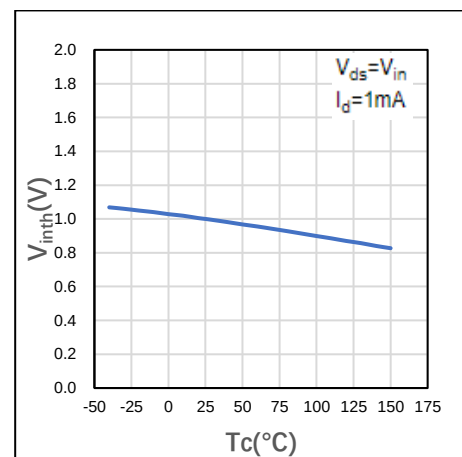


Figure 2-26 Normalized input threshold voltage vs temperature

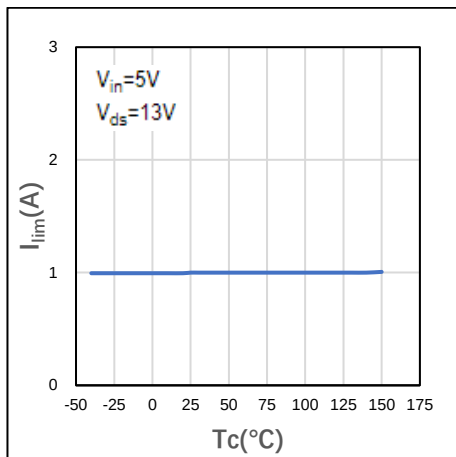


Figure 2-27 Normalized current limit vs junction temperature

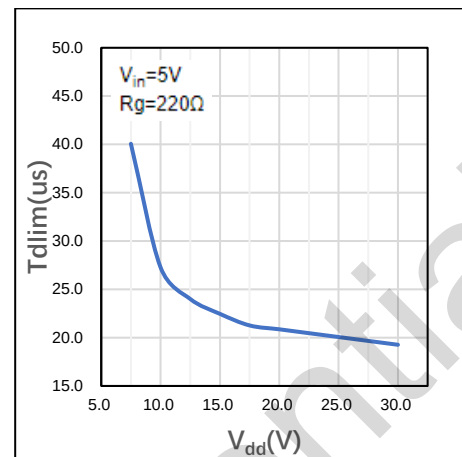


Figure 2-28 Step response current limit

3 Protection features

During normal operation, the input pin is electrically connected to the gate of the internal Power MOSFET through a low impedance path.

The device then behaves like a standard Power MOSFET and can be used as a switch. The only difference from the user's standpoint is that a DC current flows into the input pin in order to supply the internal circuitry.

The device integrates:

- Over-voltage clamp protection: internally set at 45 V, along with the rugged avalanche characteristics of the Power MOSFET stage give this device unrivalled ruggedness and energy handling capability. This feature is mainly important when driving inductive loads.
- Linear current limiter circuit: limits the drain current I_D to I_{lim} whatever the input pin voltages. When the current limiter is active, the device operates in the linear region, so power dissipation may exceed the capability of the device. Both case and junction temperatures increase, and if this phase lasts long enough, junction temperature may reach the over temperature threshold T_{jsh} .
- Over-temperature and short circuit protection: these are based on sensing the chip temperature and are not dependent on the input voltage. The location of the sensing element on the chip in the power stage area ensures fast, accurate detection of the junction temperature. Over temperature cutout occurs in the range 150°C to 190°C, a typical value being 170°C. The device is automatically restarted when the chip temperature falls of about 15°C below shutdown temperature.
- Status feedback: in the case of an over temperature fault condition ($T_j > T_{jsh}$), the device tries to sink a diagnostic current I_{gf} through the input pin in order to indicate fault condition. If driven from a low impedance source, this current may be used in order to warn the control circuit of a device shutdown. If the drive impedance is high enough so that the input pin driver is not able to supply the current I_{gf} , the input pin will fall to 0 V. This will not however affect the device operation: no requirement is put on the current capability of the input pin driver except to be able to supply the normal operation drive current I_{ISS} .

Additional features of this device are ESD protection according to the Human Body model and the ability to be driven from a TTL logic circuit.

4 Package and packing information

4.1 SOT223 mechanical data

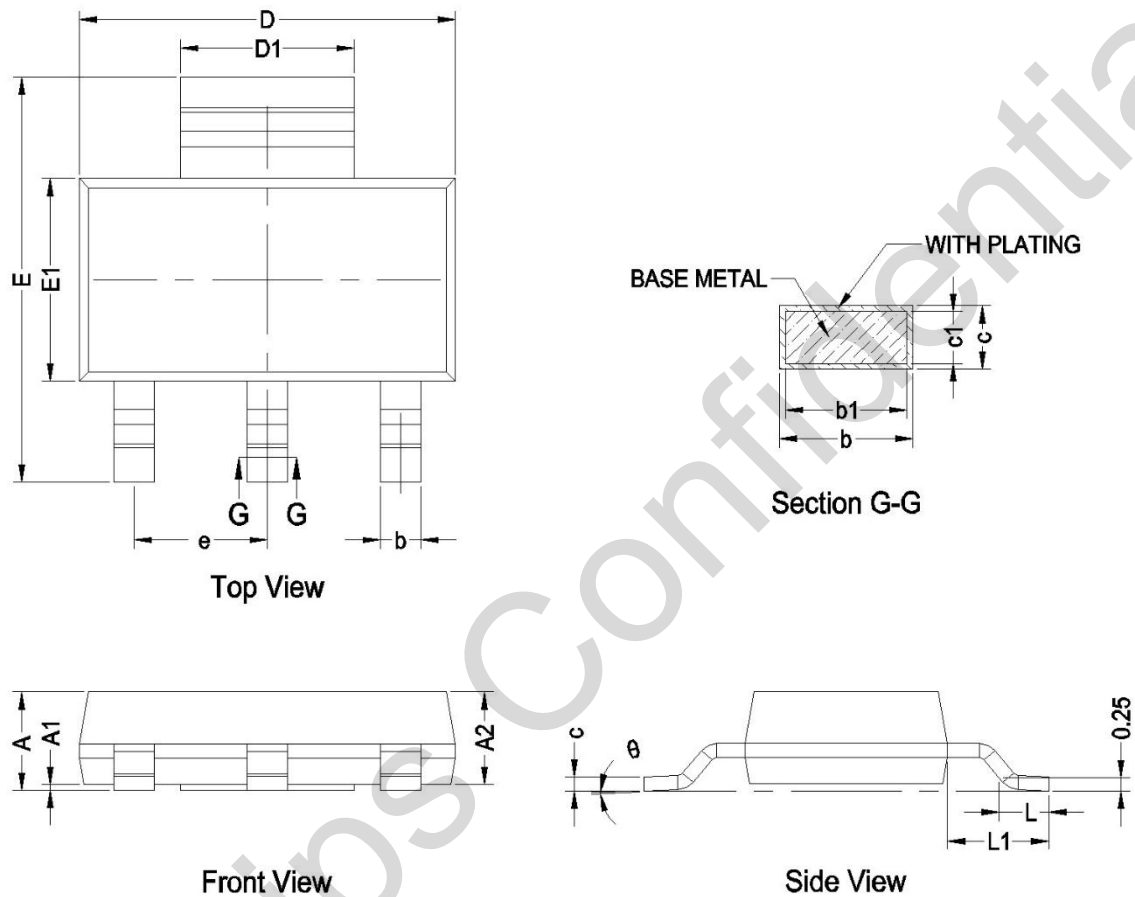


Figure 4-1 SOT223 package dimensions

Table 4-1 SOT223 mechanical data

Dim.	mm			Description
	Min.	Typ.	Max.	
A	1.50	1.65	1.80	Total Thickness
A1	0.03	0.06	0.09	Stand Off
A2	1.50	1.60	1.70	Mold Thickness
D1	2.90	3.00	3.10	Lead Width(Plating)
b	0.69	-	0.77	Lead Width(Plating)
b1	0.68	0.71	0.74	Lead Width
c	0.30	-	0.34	L/F Thickness(Plating)
c1	0.30REF	-	-	L/F Thickness
D	6.40	6.50	6.60	Body Size(Length)
E	6.80	7.00	7.20	Lead Tip to Tip
E1	3.40	3.50	3.60	Body Size(Width)
e	2.30 BSC			Lead Pitch
L	0.90	-	1.15	Foot Length
θ	0°	-	8°	Foot Angle

5 Recommended furnace temperature curve

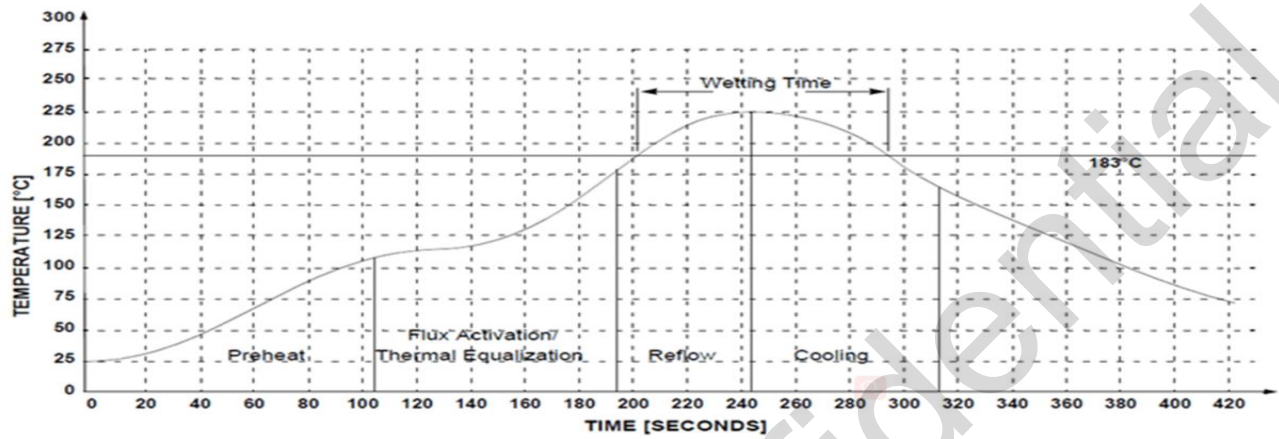


Figure 5-1 Recommended furnace temperature curve

6 Recommended pad size

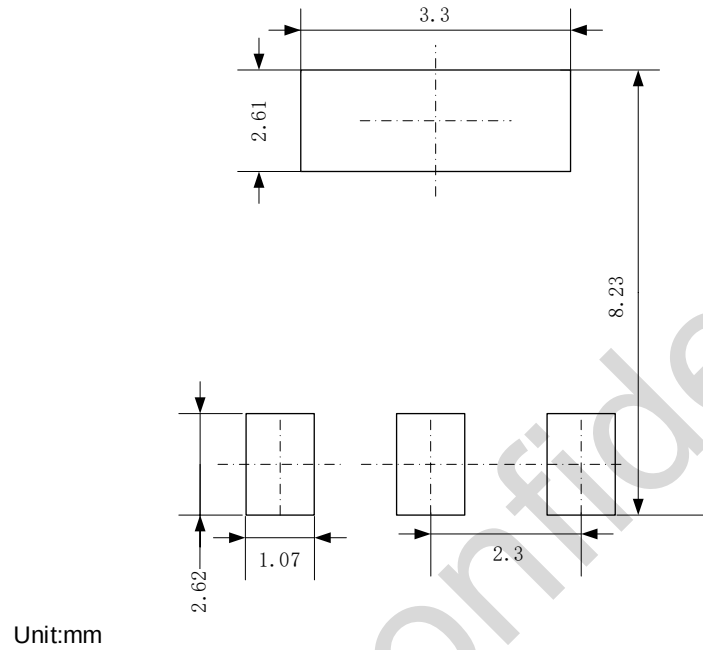


Figure 6-1 Recommended pad size

7 Customer Service

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Revision history

Date	Revision	Changes
2024/6/6	0.1.1	1.Create document version 0.1.1
2024/6/7	0.1.2	1.Change the value of Ilim
2024/7/4	0.1.3	1.Changed the format of the curve in Section 2.4
2024/7/12	1.0	1.Parameter confirmation, version upgrade to 1.0
2024/9/3	1.1	1.Updated the company's new address
2024/9/12	1.2	1.Update part number.
2024/12/21	1.3	1.Format change and update figure 2-3 2.Revise the description of Protection features
2025/1/17	1.4	1.Revised cover description. 2.Revised 3 Protection features description
2025/3/10	1.5	1. Correct error information in Figure 6-1.